

# Mohamad Abboud

FULL-STACK SOFTWARE ENGINEER

mohammad.a.abboud@gmail.com · linkedin.com/in/mohamad-abboud-97088a220 · github.com/MoAbboud · moabboud.dev

Five years of end-to-end ownership of client-facing applications, from requirements and architecture through deployment, infrastructure and ongoing support. C#/.NET, Python and PHP, across secure financial transaction platforms and a portfolio of commercial web products, as the sole technical owner and the primary client contact. Since 2026, designing, training and evaluating my own machine-learning systems end to end, without hosted AI APIs.

|                            |                             |                           |                         |
|----------------------------|-----------------------------|---------------------------|-------------------------|
| 5                          | 400+                        | 6                         | 80%                     |
| YEARS END-TO-END OWNERSHIP | BANKING INSTITUTIONS SERVED | CLIENT APPLICATIONS OWNED | PROCESSING TIME REMOVED |

01 EXPERIENCE

Independent Projects SOFTWARE ENGINEER Jan 2026 - Present · Kansas City, MO

- Designed, trained and evaluated my own machine-learning models end to end in PyTorch, with no hosted AI APIs: a **two-stage classifier** (a DistilRoBERTa detector, then a DistilBERT typer across 13 fallacy types) that flags passages which may lean on a logical fallacy, with a confidence score rather than a verdict (**Fallacy Detector**).
- Built a document-intake pipeline that turns messy invoices into validated PostgreSQL records, with a review queue for the documents it is unsure about. Its own evaluation reports **98.3% field accuracy** against a 75.1% baseline, covered by **355 tests (Mailman)**.
- Built portable memory for AI chat: a long conversation becomes a compact brief where every line traces back to the messages it came from, benchmarked against a plain summary at the same token budget on a **261-fact answer key** written by hand (**Herder**).
- Wrote a WebGL2 renderer from scratch, with no engine, for a voxel diorama a camera walks through on a timeline, for recording video, covered by **620 tests (Trail)**.
- Run this site end to end: requirements written before the code, an allowlist build published by **GitHub Actions** on every push, and **Docker Compose with Caddy** for the services that run on a server, on my own domain.

Tekkii FULL STACK WEB DEVELOPER Jan 2023 - Jan 2026 · Kansas City, KS

- Owned a portfolio of **six client web applications** across logistics, e-commerce, public sector and retail as the sole technical lead, driving each engagement from requirements and client communication through architecture, development, deployment and ongoing support in an Agile/Scrum process.
- Architected and built an automated **EDI capture and transmission system** from the ground up for Kansas City Drayage, replacing an entirely manual data-entry workflow, cutting processing time by **roughly 80%** and contributing to a **50% increase in profit**.
- Set the backend direction across the client portfolio, designing RESTful APIs, service layers and optimized SQL databases (PostgreSQL, SQL Server, MySQL) in Python and PHP, and integrating third-party services including PayPal, SendGrid and OneSignal.
- Inherited and stabilized applications built by previous developers, clearing long-standing defects and advising application owners on architecture, technical debt and feature direction.
- Led the rebuild of an e-commerce platform's backend and database layer, driving growth in traffic and revenue that supported the product's **successful acquisition**.
- Delivered an interactive, hardware-integrated installation at the **College Basketball Experience**, personally integrating a custom application with cameras, displays and a teleprompter.

Allied Engineering Group SOFTWARE ENGINEER Jun 2019 - Jun 2021 · Beirut, Lebanon

- Built and maintained **SWIFT financial transaction platforms** in C#/ASP.NET on SQL Server, processing **thousands of secure transactions daily** for over **400 banking institutions** worldwide, including QNB.

- Owned the data layer for mission-critical, high-throughput transaction systems, designing schemas and stored procedures and optimizing complex queries to improve application performance and reliability.
- Served as the direct technical liaison to banking clients, gathering requirements and delivering customized interfaces and features that measurably increased usability and client satisfaction.
- Diagnosed and resolved production issues by monitoring SQL and application logs, maintaining uninterrupted operations for banking clients and surfacing risks and blockers ahead of release.

## 02 CAPABILITIES

---

|                            |                                                                                                                                                                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01 LANGUAGES               | PHP, Python, JavaScript, C#, SQL                                                                                                                                                                                                           |
| 02 WEB DEVELOPMENT         | Public-Facing Web Applications, REST APIs, HTML / CSS, Vue.js, React, Angular, Node.js, Responsive Design                                                                                                                                  |
| 03 DATABASES               | PostgreSQL, SQL Server, MySQL, Query Optimization, Stored Procedures, Schema Design, Data Modeling, Large Dataset Processing                                                                                                               |
| 04 ARCHITECTURE & DELIVERY | System Design, API Design, Legacy System Modernization, Web Application Security, Authentication, Git / GitHub, Docker, CI/CD Pipelines, Server Administration (SSH/FTP), Agile / Scrum, Requirements Gathering, Stakeholder Communication |
| 05 MACHINE LEARNING        | PyTorch, Fine-Tuning Transformers, Deep Learning, Model Evaluation, Local Models, No Hosted APIs, Vector Search                                                                                                                            |

## 03 PROJECTS

---

WEBGL2 · NO ENGINE

### Trail

A voxel diorama a camera walks through on a timeline, for recording video. Hand-written WebGL2, no engine, 620 tests.

PYTHON · PYTORCH · FLASK

### Fallacy Detector

Flags passages that may lean on a logical fallacy, as a warning with a confidence, never a verdict. A DistilRoBERTa detector and a DistilBERT typer across 13 fallacy types, trained for the job; no hosted model, no API key.

PYTHON · DEEP LEARNING

### Pneumonia Detection

Deep-learning application classifying chest X-rays for signs of pneumonia. Free hosting, so the first load takes about a minute to wake.

FASTAPI · POSTGRESQL · DOCKER

### Mailman

Document intake: messy invoices in, validated records out, with a review queue for the ones it is unsure about. Its own evaluation reports 98.3% field accuracy against a 75.1% baseline; 355 tests.

FASTAPI · PGVECTOR · LOCAL MODELS

### Herder

● **IN PROGRESS: TESTING AND A LIVE DEMO STILL TO COME**

Portable memory for AI chat: a long conversation becomes a compact brief where every line traces back to the messages it came from. Benchmarked against a plain summary at the same token budget, on a 261-fact answer key written by hand.

PYTHON · LLM TOOLING

### Evaluators

Two AI-engineering pieces: a harness that scores structured extraction case by case, and typed GitHub tools for a triage agent.

## 04 EDUCATION

---

UNIVERSITY OF MISSOURI-KANSAS CITY

### Master of Science, Computer Science

2021 - 2023

**Pneumonia Detection App:** A deep learning application for medical image analysis in Python, classifying chest X-rays for signs of pneumonia.